DataLift Business Enablement & Services LLC
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BUILDING AN END TO END RETAIL ANALYTICS PLATFORM

From raw CSV files to executive dashboard reporting using a Medallion Lakehouse Architecture

NAME OF PROJECT:
Retail Sales & Inventory Analysis

PRESENTED BY:
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Agenda

Business Problem

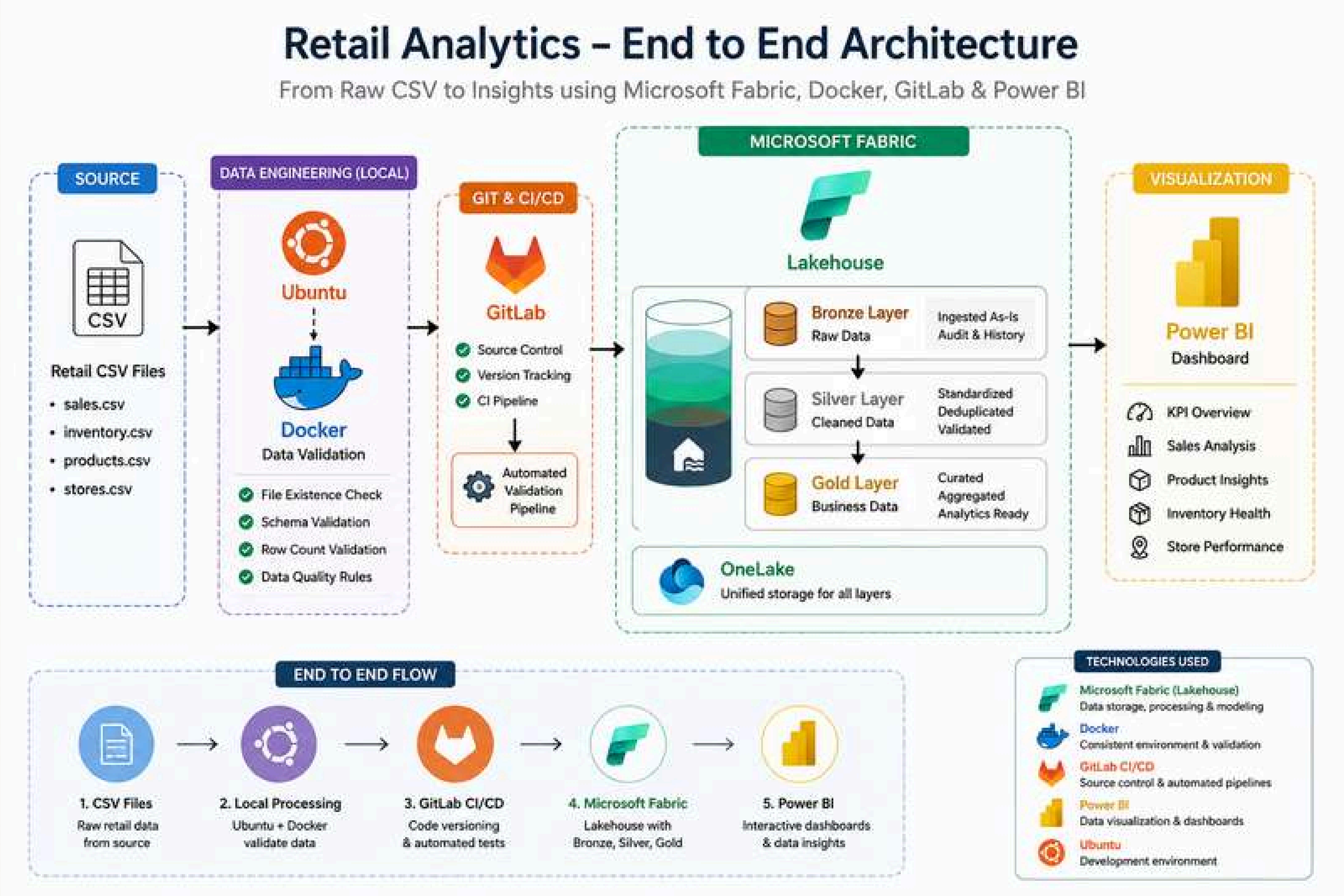
Problem Statement

Retail organizations often struggle with:

- siloed sales and inventory data
- inconsistent reporting
- lack of inventory visibility
- delayed analytics workflows

This project simulates a modern cloud data engineering solution that centralizes ingestion, cleansing, modeling, and reporting.

Architecture Diagram



Bronze Layer

Raw Data Ingestion

Completion Process:

- uploaded CSV files into Fabric Lakehouse
- raw ingestion strategy
- validation checks
- medallion architecture

Silver Layer

Data Cleansing & Standardization

Completion Process:

- PySpark notebook transformations
- schema enforcement
- deduplication
- derived metrics
- quality validation

Gold Layer

Analytics Ready Modeling

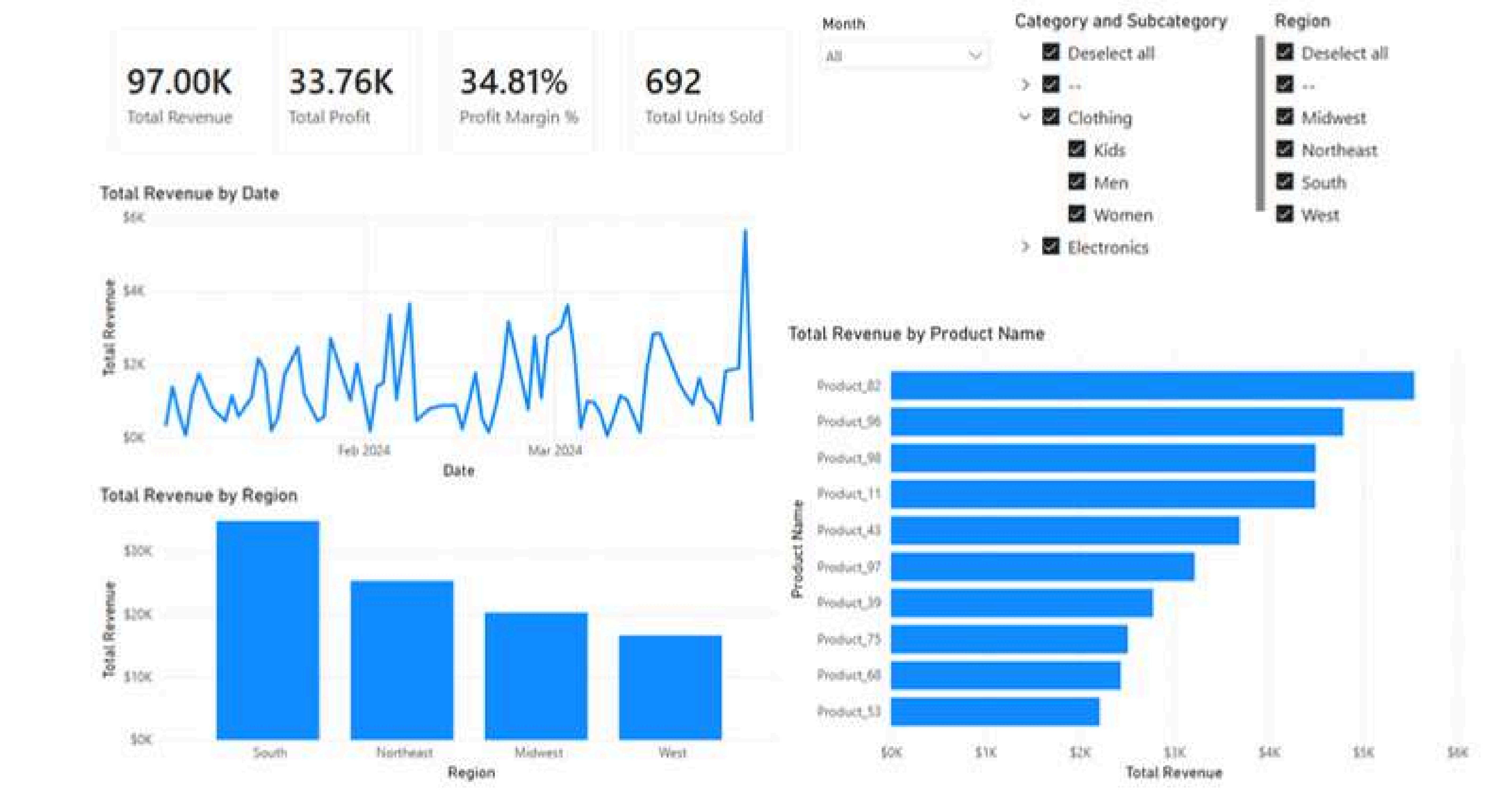
Completion Process:

- fact and dimension modeling
- semantic layer preparation
- KPI optimization

Designed for Power BI semantic modeling and executive reporting

Executive Sales & Retail Performance Dashboard

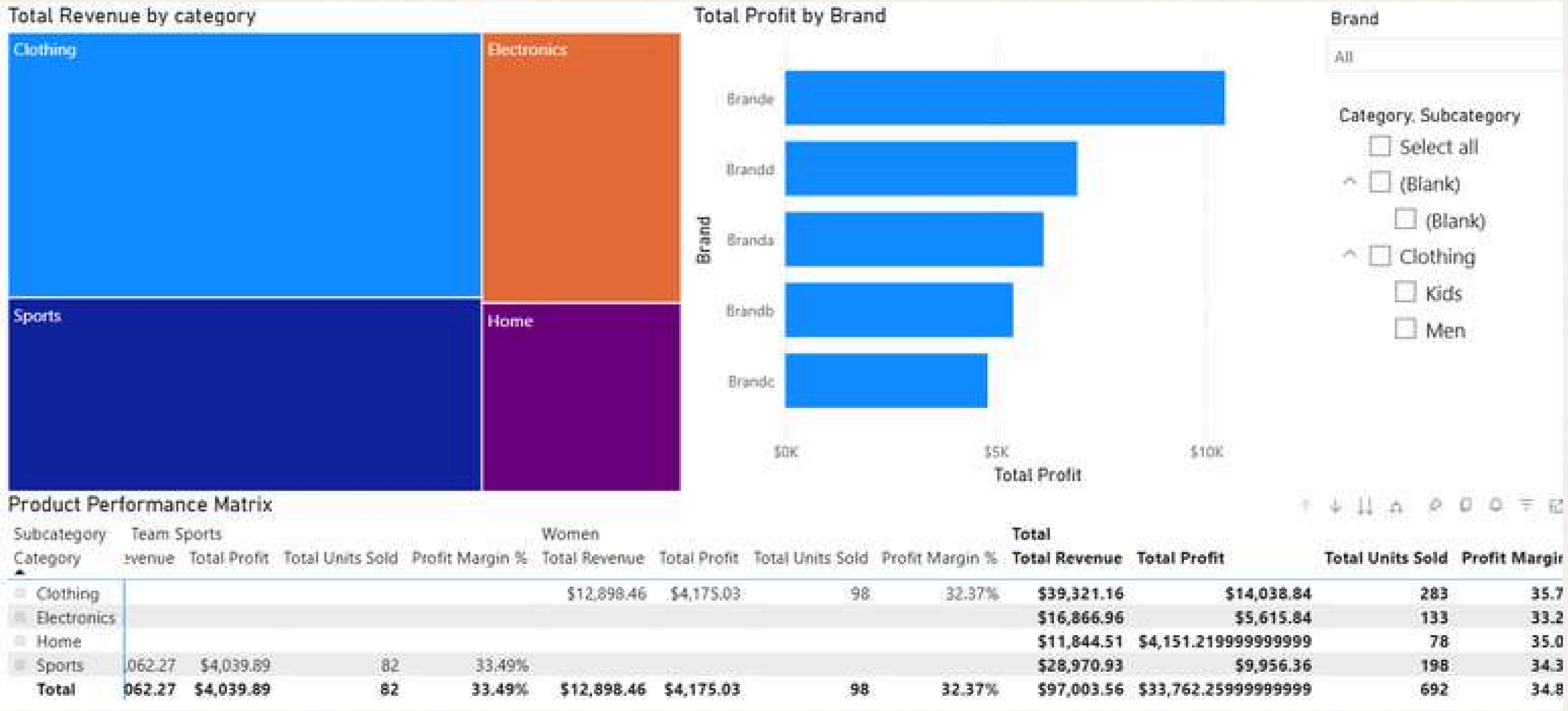
Interactive Power BI dashboard built using Microsoft Fabric Gold Layer tables and semantic modeling.



- Developed a Power BI executive dashboard connected to Microsoft Fabric Lakehouse Gold tables.
- Modeled semantic relationships between fact and dimension tables for enterprise-style reporting.
- Built KPI cards, trend analysis, regional performance metrics, and product ranking visualizations.
- Enabled self-service analytics using interactive slicers and dynamic filtering capabilities.
- Applied medallion architecture principles (Bronze → Silver → Gold) to support trusted reporting.
- Integrated curated business-ready datasets optimized for analytics consumption.

Product Performance & Profitability Dashboard

Interactive product analytics dashboard built using Microsoft Fabric Gold Layer sales tables and semantic modeling.



Subcategory / Category	Team Sports: evenue	Team Sports: Total Profit	Team Sports: Total Units Sold	Team Sports: Profit Margin %	Women: Total Revenue	Women: Total Profit	Women: Total Units Sold	Women: Profit Margin %	Total: Total Revenue	Total: Total Profit	Total: Total Units Sold	Total: Profit Margir
Clothing					$12,898.46	$4,175.03	98	32.37%	$39,321.16	$14,038.84	283	35.7
Electronics									$16,866.96	$5,615.84	133	33.2
Home									$11,844.51	$4,151.219999999999	78	35.0
Sports	,062.27	$4,039.89	82	33.49%					$28,970.93	$9,956.36	198	34.3
Total	**062.27**	**$4,039.89**	**82**	**33.49%**	**$12,898.46**	**$4,175.03**	**98**	**32.37%**	**$97,003.56**	**$33,762.25999999999**	**692**	**34.8**

- Built a product performance dashboard using Microsoft Fabric and Power BI semantic modeling.
- Designed analytics for category-level revenue, brand profitability, and product performance tracking.
- Enabled interactive drill-down analysis across categories, subcategories, and brands.
- Identified top-performing product categories and highest profit-generating brands through dynamic reporting.
- Leveraged Gold layer fact and dimension tables to support scalable business analytics.
- Delivered centralized product insights for revenue optimization and executive decision-making.
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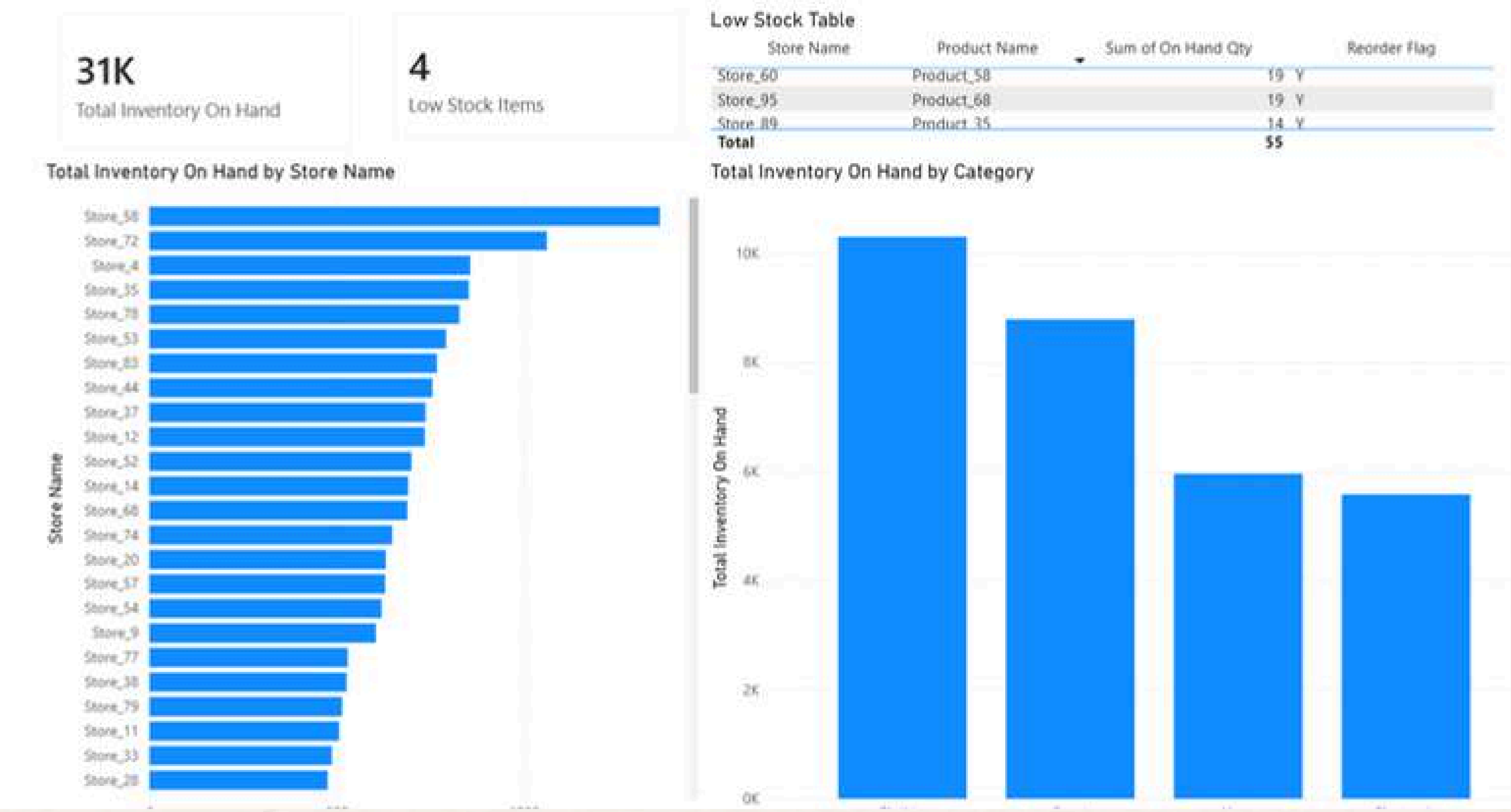
Inventory Health & Stock Monitoring Dashboard

Interactive inventory monitoring dashboard built using Microsoft Fabric Gold Layer inventory tables and semantic modeling.



- Built an inventory health dashboard using Microsoft Fabric and Power BI semantic modeling.
- Designed KPI tracking for total inventory on hand and low stock item monitoring.
- Enabled operational visibility into inventory distribution across stores and product categories.
- Identified low stock products requiring replenishment through dynamic filtering and inventory alerts.
- Leveraged Gold layer inventory fact tables for optimized reporting and analytics performance.
- Supported inventory management decision-making through centralized business-ready reporting.

Key Takeaways

Summary

- End-to-end cloud data engineering using Microsoft Fabric and medallion architecture.
- Centralized ingestion, cleansing, and modeling for trusted analytics.
- Executive dashboards built on semantic models for self-service reporting.
- Scalable, repeatable patterns applicable to real-world retail environments.
- Data-driven decision-making enabled through modern tooling and best practices.